3 Points Each

2003

1. Which of the following numbers is the greatest?

(A) 2 + 0 + 0 + 3
(B) 2 x 0 x 0 × 3
(C) (2 + 0) x (0 + 3)
(D) 20 x 0 x 3
(E) (2 x 0) + (0 x 3)

2. Zosia is drawing flowers of different colors. The first flower is blue, the second white, the next one red, the next one yellow, and again blue, white, red, yellow, and so on in the same order. What is the color of the twenty ninth flower drawn by Zosia?

(A) blue
(B) white
(C) red
(D) pink
(E) yellow

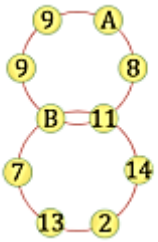
3. How many integers are there on the number line between the numbers 2.09 and 15.3?

(A) 13
(B) 14
(C) 11
(D) 12
(E) Infinitely many

4. The least positive integer which is divisible by 2, 3, and 4 is:

(A) 1
(B) 6
(C) 12
(D) 24
(E) 36

5. Two of the numbers located on the two circles (see the picture) are represented by letters A and B. The sum of the numbers on each circle is equal to 55. What number is represented by the letter A?

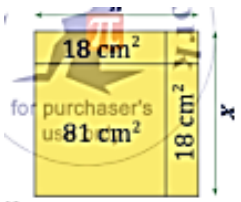


(A) 9
(B) 10
(C) 13
(D) 16
(E) 17

6. Tom has 9 bills worth 100 dollars each, 9 bills worth 10 dollars each, and 10 coins worth 1 dollar each. How much money does Tom have?

(A) $1,000
(B) $991
(C) $9,910
(D) $9,901
(E) $99,010

7. A square with the length of its side equal to x is made up of a square with an area of 81 cm², two rectangles with areas of 18 cm² each, and a small square. What is the value of x?



(A) 2 cm
(B) 7 cm
(C) 9 cm
(D) 10 cm
(E) 11 cm

8. The value of the expression $\frac{2003+2003+2003+2003+2003}{2003+2003}$ is equal to:

(A) 2003
(B) $\frac{1}{3}$
(C) 3
(D) $\frac{5}{2}$
(E) 6009

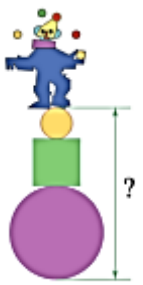
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9. Dasia likes to add the digits that indicate the current time on her electronic watch. For example, when the watch shows 21:17, she gets a sum equal to 11. What is the greatest sum she can get?

(Hint: in some countries and sometimes in the USA, instead of saying, "It is 1 p.m.," people may say, "It is 13:00." When it is 2 p.m., they may say, "It is 14:00," and when it is 12 a.m., they may say, "It is 24:00." In this problem 21:17 means 9:17 p.m. Time expressed using this method is sometimes called military time.)

(A) 24
(B) 36
(C) 19
(D) 25
(E) 28

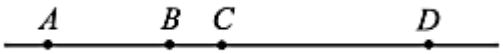
10. The picture shows Clown Ian dancing on top of two balls and a cube. The radius of the lower ball is 6 dm, and the radius of the upper ball is three times shorter. The edge of the cube is 4 dm longer than the radius of the upper ball. At what height is Ian dancing?



(A) 14 dm
(B) 20 dm
(C) 22 dm
(D) 24 dm
(E) 28 dm

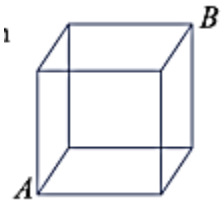
4 Points Each

11. Let AC = 10 m, BD = 15 m, AD = 22 m (see the picture below). The length of segment BC is equal to



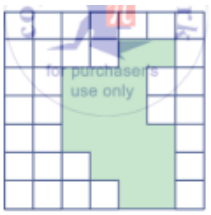
(A) 1 m
(B) 2 m
(C) 3 m
(D) 4 m
(E) 5 m

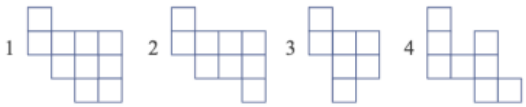
12. How many shortest distances along the edges of the cube are there that connect vertex A with the opposite vertex B?



(A) 4
(B) 6
(C) 3
(D) 12
(E) 16

13. Two pieces are cut out from a square puzzle. These two pieces made up the shaded region (see the picture). From the four pictures below, which two pieces are they?





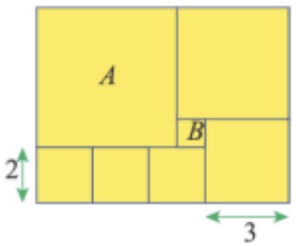
(A) 1 and 4
(B) 2 and 4
(C) 2 and 3
(D) 1 and 3
(E) 3 and 4

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14. We add together two different numbers chosen from the numbers 1, 2, 3, 4, and 5. How many different sums can we get?

(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

15. The figure in the picture consists of 7 squares. Square A has the greatest area and square B has the smallest area. The lengths of the sides of two squares are given. How many squares identical to square B would it take to fill in square A completely?



(A) 16
(B) 25
(C) 36
(D) 49
(E) It is impossible.

16. A certain bar code consists of 17 black bars. A white bar is placed between any two black bars. The first bar and the last bar in the code are black. There are two kinds of black bars: wide and narrow. The number of white bars is 3 more than the number of wide black bars. How many narrow black bars are there in this bar code?



(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

17. Eva has 20 marbles of four colors: yellow, green, blue, and black. 17 of them are not green, 5 are black, and 12 are not yellow. How many blue marbles does Eva have?

(A) 3
(B) 4
(C) 6
(D) 7
(E) 8

18. There are 17 trees on one side of the street on Tom's way from his house to school. One day Tom marked these trees with white chalk in the following way: on the way from his house to school he marked every other tree, starting with the first one, and on his way back home he marked every third tree, starting with the first one. How many trees were not marked?

(A) 4
(B) 5
(C) 6
(D) 7
(E) 8

19. Today's date is written as 3.21.2003 (March 21st, 2003) and the time is 20:03 (8:03 p.m.). How will the date be written after 2003 minutes?

(A) 3.21.2003
(B) 3.22.2003
(C) 3.23.2003
(D) 4.21.2003
(E) 4.22.2003

20. What is the ones digit in the number 2003^{2003}?

(A) 7
(B) 1
(C) 9
(D) 5
(E) 3

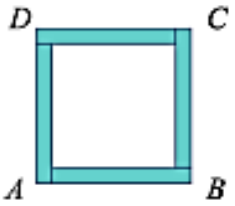
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5 Points Each

21. With how many zeros does the product of the consecutive natural numbers from 1 to 50 end?

(A) 5
(B) 10
(C) 12
(D) 20
(E) 50

22. The square ABCD consists of a white square and four shaded The area of square ABCD equals: rectangles. Each of the shaded rectangles has a perimeter of 40 cm. The area of square ABCD equals:

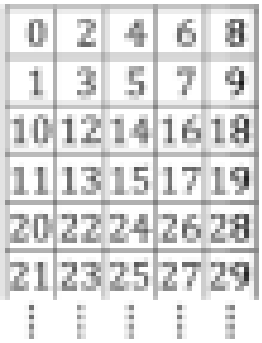


(A) 100 cm²
(B) 200 cm²
(C) 160 cm²
(D) 400 cm²
(E) 80 cm²

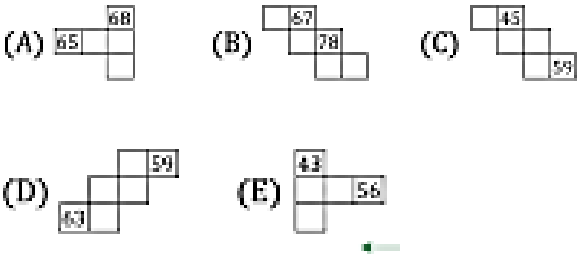
23. We have six segments with lengths of 1, 2, 3, 2001, 2002, and 2003. In how many ways can we select three of these segments to build a triangle?

(A) 1
(B) 3
(C) 5
(D) 6
(E) 10

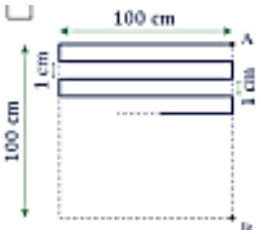
24. Pete is writing the numbers from 0 to 109 into a five-column table using a rule which is easy to figure out (see the picture to the right). Which of the pieces below cannot be filled in with numbers to fit Pete's table?



0	2	4	6	8
1	3	5	7	9
10	12	14	16	18
11	13	15	17	19
20	22	24	26	28
21	23	25	27	29



25. In the figure, the beginning part of the path from point A to point B is shown. How long is the whole path?



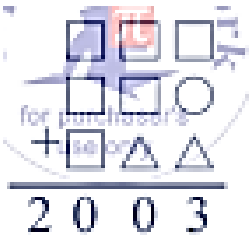
(A) 10,200 cm
(D) 10,100 cm
(B) 2,500 cm
(E) 9,900 cm
(C) 909 cm

26. At 3:00 o'clock the minute hand and the hour hand make a right angle. What will the measure of the angle between these hands be after 10 minutes?

(A) 90°
(B) 30°
(C) 80°
(D) 60°
(E) 35°

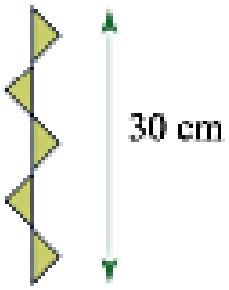
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27. In the addition problem shown in the picture, every square stands for a certain digit, every triangle stands for another digit, and every circle denotes yet another digit. What is the sum of the numbers represented by the square and the circle?



(A) 6
(B) 7
(C) 8
(D) 9
(E) 13

28. The shaded figure in the picture to the right consists of five identical isosceles right triangles. The area of the shaded figure is:



(A) 20 cm^2
(B) 25 cm^2
(C) 35 cm^2
(D) 45 cm^2
(E) 60 cm^2

29. Red and green dragons lived in a cave. Each red dragon had 6 heads, 8 legs, and 2 tails. Each green dragon had 8 heads, 6 legs, and 4 tails. There were 44 tails altogether, and there were 6 fewer green legs than red heads. How many red dragons lived in the cave?

(A) 6
(B) 7
(C) 8
(D) 9
(E) 10

30. Anya has 9 crayons in a box. At least one of them is blue. At least two of every 4 crayons are of the same color, and at most three out of every 5 crayons are of the same color. How many blue crayons are in this box?

(A) 2
(B) 3
(C) 4
(D) 1
(E) 5

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Answers key:

1. C
2. A
3. A
4. C
5. B
6. A
7. E
8. D
9. A
10. C
11. C
12. B
13. D
14. C
15. B
16. D
17. B
18. B
19. C
20. A
21. C
22. D
23. D
24. B
25. D
26. E
27. A
28. D
29. C
30. B

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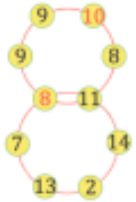
1. (C) $(2 + 0) \times (0 + 3)$
One factor equal to 0 makes the whole product equal to 0 and adding 0 to a number doesn't change the number, so $2 \times 0 \times 0 \times 3$, $20 \times 0 \times 3$, and $(2 \times 0) + (0 \times 3)$ are all equal to 0. $2 + 0 + 0 + 3 = 5$ and $(2 + 0) \times (0 + 3) = 2 \times 3 = 6$, so $(2 + 0) \times (0 + 3)$ is the greatest number among the five given numbers.

2. (A) blue
Zosia is using 4 different colors. $29 = 7 \times 4 + 1$, so the twenty ninth flower drawn by Zosia has the same color as the first one, which is blue.

3. (A) 13
Integers between 2.09 and 15.3 are 3, 4, 5, ..., 14, 15. There are $15 - 2 = 13$ of them.

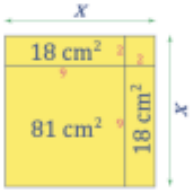
4. (C) 12
The least positive integer divisible by 2 and 3 is 6. 6 is not divisible by 4. The next multiple of 6 is 12, and 12 is the least positive integer divisible by 2, 3, and 4.

5. (B) 10
$B + 9 + 9 + A + 8 + 11 = 55$ and $B + 11 + 14 + 2 + 13 + 7 = 55$, so $B + A + 37 = B + 47$. Hence, $A = 10$. Notice that for computing A we only have to know that $B + 9 + 9 + A + 8 + 11 = B + 11 + 14 + 2 + 13 + 7$.
If both sums are equal to 55, then $B = 8$.



6. (A) \$1,000
$9 \times \$100 + 9 \times \$10 + 10 \times \$1 = \$900 + \$90 + \$10 = \$1000$, so Tom has \$1,000.

7. (E) 11 cm



Each side of the large square is 9 cm since $(9 \text{ cm})^2 = 81 \text{ cm}^2$.
The rectangle above the large square has one side equal to 9 cm and an area of 18 cm², so the vertical side of the rectangle is equal to 2 cm since $2 \text{ cm} \times 9 \text{ cm} = 18 \text{ cm}^2$. Therefore, the left side of the big square is 9 cm + 2 cm, which is 11 cm. Hence, $x = 11$ cm.
Notice that the area of the small square at the top right corner is 4 cm² and $81 + 2 \times 18 + 4 = 9^2 + 2 \times 9 \times 2 + 2^2 = (9 + 2)^2 = 11^2$.

8. (D) $\frac{5}{2}$

$$\frac{2003 + 2003 + 2003 + 2003 + 2003}{2003 + 2003} = \frac{5 \times 2003}{2 \times 2003} = \frac{5}{2}$$

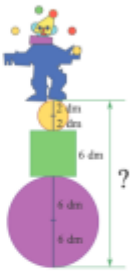
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9. (A) 24

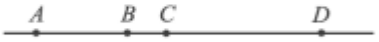
The first minute after midnight is reported as 00:01. Then 00:02, ...,00:59, 01:00, 01:01, ... 01:59, 02:00, 02:01,..., 02:59, 3:00, ..., 23:00, 23:01,..., 23:59.
After that, the time is either 24:00, 00:01, 00:02,... or 00:00, 00:01, 00:02,... . There is no 24:01 or 24:59. The sum of digits for 24:00 is 6 and the sum of digits for 23:59 is 19 but 19 is not the greatest sum. It is true that $5 + 9 = 14$ is the greatest minute sum but the greatest hour sum is $1 + 9 = 10$, so the greatest sum of all four digits is $10 + 14 = 24$. It is the sum for 19:59.

10. (C) 22 dm

A radius is a segment from the center of a circle or sphere to its circumference or surface. The diameter is the maximum distance between points on the circumference or surface.
The radius of the lower ball is 6 dm and the radius of the upper ball is 2 dm since $6 \div 3 = 2$. The edge of the cube is 2 dm + 4 dm = 6 dm. Ian is dancing at a height which is the sum of the diameters of both balls and the edge of the cube. $2 \times 6 + 6 + 2 \times 2 = 22$, so Ian is dancing at the height of 22 dm.

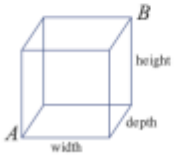


11. (C) 3 m



AD is 22 meters long and BD is 15 meters long, so AB is $22 - 15 = 7$ meters long.
AC is 10 meters long and AB is 7 meters long, so BC is $10 - 7 = 3$ meters long.
Hence, the length of segment BC is equal to 3 m.

12. (B) 6

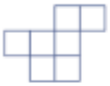


From each vertex we can move in 3 directions: width, depth, height. The shortest path consists of three consecutive edges in 3 different directions selected in any order. There are 6 such options: w-d-h, w-h-d, d-w-h, d-h-w, h-w-d, and h-d-w.

13. (D) 1 and 3

The shaded areas is made up of 17 small squares.
Pieces 1 and 2 together cover $10 + 9 = 19$ small squares, so they cannot fit in the shaded area. Pieces 1 and 4 together cover $10 + 8 = 18$ small squares, so they cannot fit in the shaded area. Pieces 2 and 3 together cover $9 + 7 = 16$ small squares, so they cannot fill the shaded area. Also, pieces 3 and 4 together cover $7 + 8 = 15$ small squares, so they cannot fill the shaded area. The only pairs of pieces that cover an area are pieces 1 and 3 ($10 + 7 = 17$) and pieces 2 and 4 ($9 + 8 = 17$).





After rotating 90° clockwise (as seen to the left), piece 3 fits in the green area and piece 1 fits directly in the red area.

The only two ways that piece 4 can fit into the shaded region is shown in orange below. There is no possibility to fit any other pieces to cover the remaining shaded area.



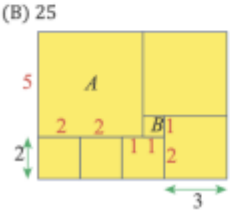


Therefore, only the combination of Pieces 1 and 3 is the right fit for the shaded region.

14. (C) 7

The smallest sum is $1 + 2 = 3$ and the largest sum is $4 + 5 = 9$. There are 7 integers from 3 to 9, and each of them is one of the sums. Indeed, $4 = 1 + 3$, $5 = 1 + 4$, $6 = 1 + 5$, $7 = 2 + 5$, and $8 = 3 + 5$. Hence, we can get 7 different sums.

15. (B) 25



The difference between the left edge of 3×3 square (the bottom right corner of the big rectangle) and the right edge of 2×2 square adjacent to it is 1, so B is a 1×1 square. Hence the bottom edge of the square A is $2 + 2 + 2 - 1 = 5$. A is a 5×5 square and B is a 1×1 square, so we need 25 B-squares to fill in square A completely.

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16. (D) 4
There are 16 white bars since the 1st bar is black and the 17th bar is black. There are 13 wide black bars since 16 – 3 = 13. Also, 17 – 13 = 4, so there are 4 narrow black bars in this bar code.

17. (B) 4
Eva has 3 green marbles since 20 – 17 = 3. She has 8 yellow marbles since 20 – 12 = 8. 5 marbles are black, so Eva has 20 – (3 + 8 + 5) = 4 blue marbles.

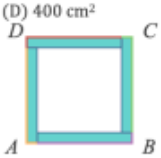
18. (B) 5
Enumerate consecutive trees from Tom's house to school as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17. When he walks to school, he marks trees 1, 3, 5, 7, 9, 11, 13, 15, 17. When he walks back, he marks trees 17, 14, 11, 8, 5, and 2. The trees 4, 6, 10, 12, and 16 are not marked, and there are 5 of them. Notice that 4, 6, 10, 12, 16 are even numbers between 1 and 17 such that (17 – even number) is not a multiple of 3.

19. (C) 3.23.2003
2003 minutes = 33 × 60 minutes + 23 minutes = 33 hours + 23 minutes = = 24 hours + 9 hours + 23 minutes = 1 day + 9 hours + 23 minutes. Today the time is 20:03, so one day later will be tomorrow (March 22nd) at 20:03; 9 hours later will be the day after tomorrow (March 23rd) at 5:03 a.m., and 23 minutes later will be the day after tomorrow at 5:26. Thus, the date after 2003 minutes will be 3.23.2003 (March 23rd, 2003, at 5:26 a.m.).

20. (A) 7
The last digit of 2003^4 is the last digit of $3 \times 3 \times 3 \times 3 = 9 \times 9 = 81$, so the last digit of 2003^4 is 1. $2003^{2003} = [(2003)^4]^{500} \times 2003^3$, so the last digit of 2003^{2003} is the last digit of 2003^3. The last digit of 2003^3 is the last digit of $3 \times 3 \times 3 = 27$, which is 7. Hence, the last digit of 2003^{2003} is 7.

21. (C) 12
5, 5 × 2, 5 × 3, 5 × 4, 5 × 5, 5 × 6, 5 × 7, 5 × 8, 5 × 9, and 5 × 5 × 2 are multiples of 5 from 1 to 50. The factor 5 is repeated 12 times in the prime factorization of the product of consecutive numbers from 1 to 50. The factor 2 is repeated more than 12 times (actually 47 times) but we only need to pair up 2s with the twelve 5s to get twelve factors of 10. Therefore, the product of consecutive numbers from 1 to 50 ends with 12 zeros.

22. (D) 400 cm²



Half of the perimeter of each shaded rectangle is 20 cm and each shaded rectangle shares half of its perimeter with the perimeter of the square *ABCD* as shown in four different colors for the four rectangles. These four halves cover exactly the perimeter of the square *ABCD*, so the perimeter of the square is 4 × 20 cm. Hence, each side of the square has the length of (4 × 20 cm) ÷ 4 = 20 cm and the area of the square *ABCD* is $(20 \text{ cm})^2$, which is 400 cm².

23. (D) 6
In any triangle the sum of the lengths of any two sides is strictly greater than the length of the third side. It is enough when the sum of two shorter lengths is greater than the longest one. This immediately excludes 1 as the shortest side.
2 can be the shortest side in two triangles since 2 + 2001 > 2002 and 2 + 2002 > 2003.
3 can be the shortest side in three triangles since 3 + 2001 > 2002, 3 + 2001 > 2003, and 3 + 2002 > 2003. Also, 2001 + 2002 > 2003. Thus, we can build 6 triangles using the given segments.

24. (B)

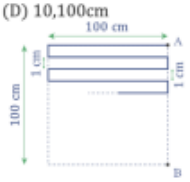


0	2	4	6	8
1	3	5	7	9
10	12	14	16	18
11	13	15	17	19
20	22	24	26	28
21	23	25	27	29

In each column the last digits are two alternating digits. The two alternating digits are: 0 and 1 in the first column, 2 and 3 in the second column, 4 and 5 in the third column, 6 and 7 in the fourth column, and 8 and 9 in the fifth column. Thus, there can be no cell to the right of the column with 8 or 9. Hence, the piece (B) cannot be a part of the table. To the right, the table is extended to confirm that the other pieces shown below are parts of the table.

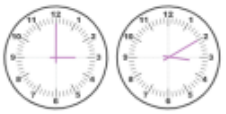
40	42	44	46	48
41	43	45	47	49
50	52	54	56	58
51	53	55	57	59
60	62	64	66	68
61	63	65	67	69
70	72	74	76	78
71	73	75	77	79

25. (D) 10,100cm



Enumerate vertical segments (dashed, solid, dashed, solid, and so on) from A to B along the right edge. The odd numbers indicate the dashed segments and the even numbers indicate the solid segments. The last segment (ending at B) is the 100th segment (100 is an even number), so it is solid. This vertical 1 cm long segment is the last segment of the path from A to B, so the last horizontal segment of the path is just 1 cm above the bottom edge. Therefore, the path contains 100 horizontal segments each 100 cm long. They add up to 100 × 100 cm = 10, 000 cm. The vertical dashed segments have their counterparts along the left edge as segments of the path, so all the vertical pieces of the path add up to 100 cm. 10,000 cm + 100 cm = 10,100 cm, so the whole path is 10,100 cm long.

26. (E) 35°



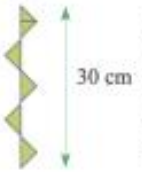
The minute hand makes a full rotation (360°) in 60 minutes, so in 10 minutes it moves 60° to 2 on the face of the clock. The angle between two consecutive numbers marked on the clock is 360° ÷ 12 = 30°. Thus, the hour hand moves 30° in 1 hour.
1 hour = 60 minutes, so during 10 minutes the hour hand moves 30° ÷ 6 = 5°.
The angle from "2" to "3" is 30°, and the hour hand is 5° beyond "3," so at 3:10 the angle between the hands is 30° + 5° = 35°.

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27. (A) 6

If you insert 5 (or less) in the squares, then you are adding 3 numbers less than 600 each. The sum cannot be 2003 since it is less than 1800. If you insert 7 (or more) in the squares, then you are adding 3 numbers greater than (or equal to) 700 each. The sum cannot be 2003 since it is at least 2100. Hence, the square represents 6. 2003 is the sum of the three numbers, so 2003 = 666 + 660 + the circle digit + 600 + the two-digit number with identical digits.
The circle digit + the two-digit number with identical digits = 2003 – (666 + 660 + 600) = = 2003 – 1926 = 77. Any two-digit number with identical digits is a multiple of 11, so the circle digit = (77 – the two-digit number with identical digits) is a multiple of 11.
The only digit that is a multiple of 11 is 0, so the circle digit is 0 and the number represented by two triangle digits is 77. 666 + 660 + 677 = 2003 and it is the only solution for this puzzle, so the sum of the numbers represented by the square and by the circle is 6 + 0 = 6.

28. (D) 45 cm^2



The hypotenuse of each triangle is equal to 30 cm ÷ 5 = 6 cm. The height of the triangle (shown) divides it into two isosceles right triangles, so the height is equal to half of the original hypotenuse. Each shaded triangle has the base equal to 6 cm and the height equal to 3 cm. The area of one shaded triangle is ½ × 6 cm × 3 cm = 9 cm^2, so the area of the shaded figure is 5 × 9 cm^2 = 45 cm^2.

29. (C) 8

Notation: r is the number of red dragons and g is the number of green dragons.
The red dragons have 6r heads, 8r legs, and 2r tails.
The green dragons have 8g heads, 6g legs, and 4g tails.
2r + 4g = 44 and 6g = 6r – 6. From the second equation g = r – 1. Substitute it into the first equation to have 2r + 4(r – 1) = 44. Simplify it to 6r = 44 + 4. 6r = 48, so r = 8.
Thus, 8 red dragons (and 7 green dragons) lived in the cave.

30. (B) 3

There are at most 3 crayon colors. Otherwise, there is a group of 4 crayons in 4 different colors which contradicts, "At least two of every 4 crayons are of the same color." There are at most 3 crayons in each color. Otherwise, there is a group of 4 crayons in one color. Add any crayon to this group to contradict, "At most three out of every 5 crayons are of the same color." This means that there are at most 3 blue crayons.
If there are only two (or fewer) blue crayons, then the total number of all crayons is at most 2 (blue) + 3 + 3 (at most two other colors; not more than 3 crayons in each color). 2 + 3 + 3 = 8 but the total number is 9. Therefore, there are exactly 3 blue crayons in the box. In fact, there are exactly 3 crayon colors and exactly 3 crayons in each color.

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